Methodology

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Introduction

The continual lowering/hoisting analysis is the preferred method performing lifting analysis from a floating vessel to and from subsea. It utilizes time domain analysis as it takes into consideration the lowering / hoisting speed of the crane in addition the irregular nature of ocean waves. This method depends upon running a number of irregular wave simulations (each using a different wave train) to obtain a statistical distribution of the maximum and minimum dynamic load (or other parameter) and then use statistical methods to determine the load (or parameter) at a specified probability value. For example, the load that has a 10% probability of non-exceedance can be calculated.

Acceptance Criteria

The goal of a lifting analysis is to determine the acceptable sea states in which a lift can be performed. To assess that, various result parameters are calculated and compared against specific acceptance criteria. Generally, this includes crane and sling loads but can include other parameters such as crane wire declination, connection force, etc. Essentially, any parameter that is available in the Orcaflex results window can be used.

For each parameter, acceptance criterion needs to be defined. The acceptance criteria is a key component to performing a lifting analysis using the RLC script. The analysis is performed for multiple significant wave heights (H_s) and wave periods (T_p/T_z). For a given $H_s$and period combination, all acceptance criteria must be met. If one is not met, that sea state is not suitable for installation.

In general, guidance for this methodology is taken from DNV-ST-N001 [4], however, the guidance provided is limited. Section 16.17.3.3 of DNV-ST-N001 [4] states:

> *"The probability of exceeding the calculated extreme characteristic load (hydrodynamic + static load or snap load) in the operation period shall not exceed 10%."*

In other words, the load that has a 1/10 probability of being exceeded. This load may be used to determine the acceptability of the lifting appliance.

To help engineers interpret the DNV requirements, TechnipFMC have prepared OED-TGR-272 Part B [6]. This document provides the acceptance for many of the components including the crane load.

As indicated above, DNV requires that load that has a 10% probability of exceedance be calculated. However, many crane systems have alarms built in to warn the Crane Operator of high loads. If the load for comparison against the acceptance criteria had a 1/10 probability of being exceeded, the crane alarms would likely be set off frequently.

Therefore, the dynamic load which has a 99% probability of non-exceedance will be selected for checking the crane capacity as referenced in OED-TGR-272 Part B [6].

Other probability criteria are also listed in OED-TGR-272 Part B [6].

Statistical Analysis

Generation of Statistical Sample Data

The lowering/hoisting analysis is performed at each H_s, T_p/T_z and heading numerous times, each time using a different wave train. This process involves defining a seed within Orcaflex (which is automatically determined by the RLC script).

By performing multiple simulations using different seeds, a statistical distribution of the maximum and minimum loads (or other parameters) can be developed. The more analysis cases performed (i.e. the more seeds analyzed) the more data available to determine the distribution parameters.

This method will result in the requirement to perform thousands of simulations. This process can be quite time consuming. This is essentially the purpose of RLC, to automate this process.

Statistical Distribution Analysis

The Gumbel distribution is a particular case of the Generalized Extreme Value distribution (Fisher-Tippett). This type of statistical distribution is found to be applicable if the distribution of the sample data is of a normal or exponential type and is commonly used to model the distribution of maxima or minima of random variables, usually related to extreme events such as rainfall or earthquakes.

The Gumbel distribution can be used to determine the probability associated with maximums and minimums of a parameter.

$$PDF_{GumbelMax}(x) = \frac{1}{\beta} e^{-\left(\frac{x-\mu}{\beta} + e^{-\left(\frac{x-\mu}{\beta}\right)}\right)} \quad (1)$$

$$CDF_{GumbelMax}(x) = e^{-e^{-\left(\frac{x-\mu}{\beta}\right)}} \quad (2)$$

$$PDF_{GumbelMin}(x) = \frac{1}{\beta} e^{\left(\frac{x-\mu}{\beta} + e^{\left(\frac{x-\mu}{\beta}\right)}\right)} \quad (3)$$

$$CDF_{GumbelMin}(x) = 1 - e^{-e^{\left(\frac{x-\mu}{\beta}\right)}} \quad (4)$$

Where:

PDF	Probability Density function
CDF	Cumulative Density Function
x	Variable (i.e. load)
β	Scale parameter (>0)
μ	Location parameter

The general maximum Gumbel distribution probability density and cumulative distribution functions are shown in Figure 1.

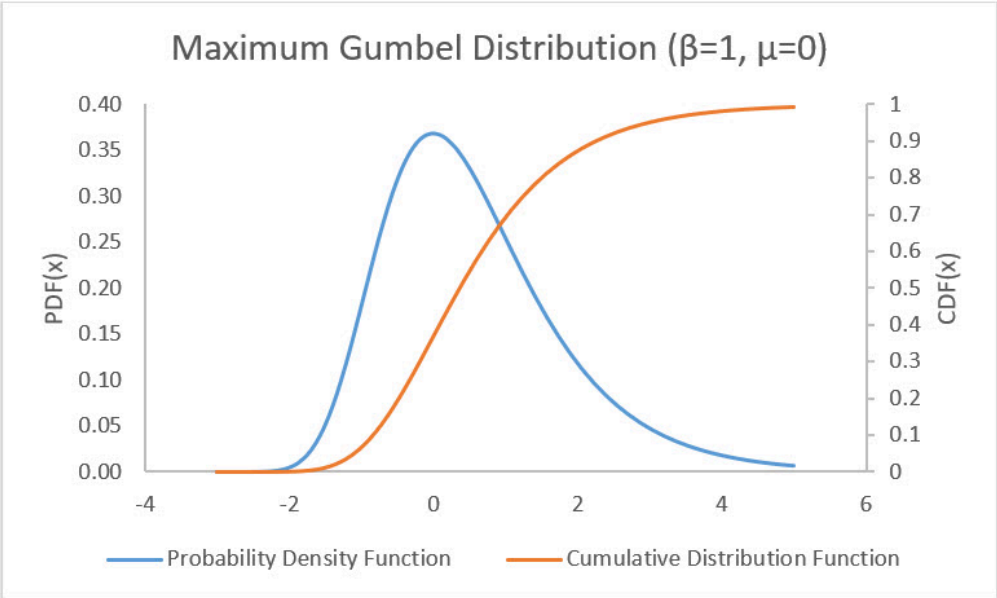


Fig. 1 *Maximum Gumbel Distribution*

The general minimum Gumbel distribution probability density and cumulative distribution functions are shown in Figure 2.

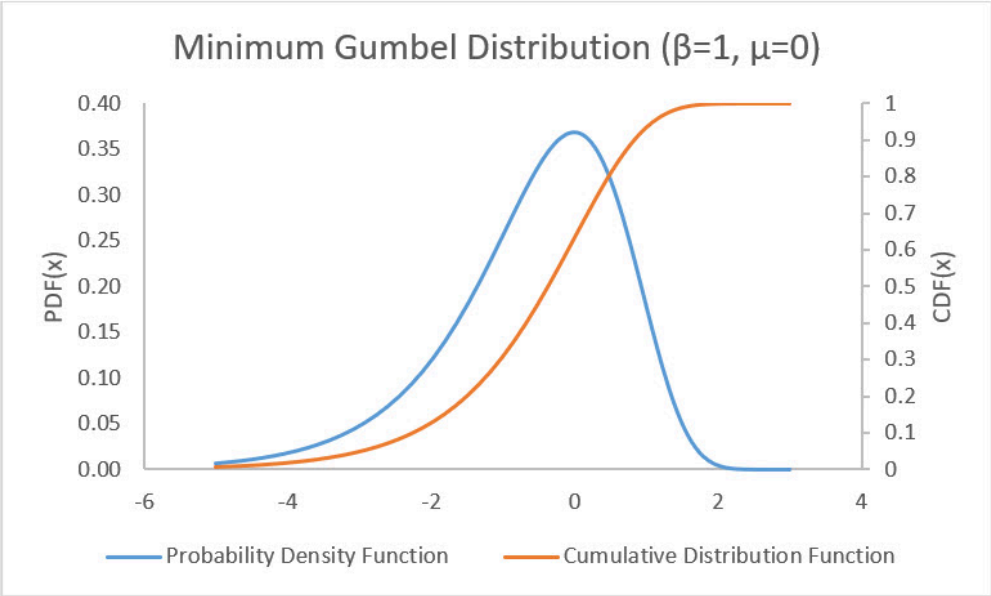


Fig. 2 *Minimum Gumbel Distribution*

It can be noted that Cumulative Density Functions (CDFs) are simply a double exponential functions. By rearranging and taking the double natural log the CDFs the equations can be rewritten.

The maximum and minimum probabilistic values of the chosen parameter can be determined:

$$Parameter_{GumbelMax}(p) = \mu - \beta \cdot ln\left(-ln(p)\right) \tag{5}$$

$$Parameter_{GumbelMin}(p) = \mu + \beta \cdot ln\left(-ln(1-p)\right) \tag{6}$$

Where:

Parameter	Dynamic Hook Load (Maximum or Minimum) with a probability of non-exceedance of “p”
p	Non-exceedance probability

These relationships can be used for any parameter of interest (i.e. sling force, crane wire angle, passive heave compensator stroke, etc.).

These formulae are in the form:

$$y = mx + b \tag{7}$$

and can therefore be plotted as a straight line.

DNV-RP-N103 [3] states:

> *“If a continuous lowering of the object is simulated a large number of realizations is required in order to obtain a proper statistical confidence.”*

It is recommended to use at least 30 different wave trains to obtain enough statistical information to reliably predict the maximum / minimum loads. Therefore, for each H_s, period and heading combination, 30 different analyses are performed, as a minimum. The maximum and minimum dynamic load is recorded for each analysis.

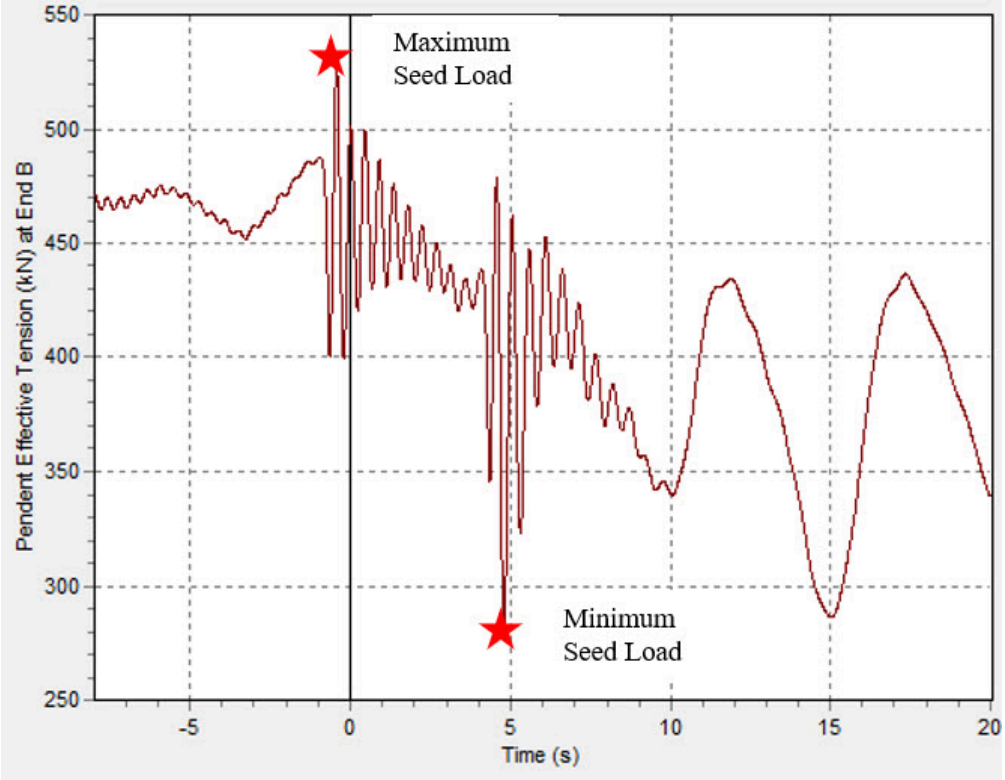


Fig. 3 *Typical Time History Plot of a Parameter (i.e. Crane Wire Tension)*

Given a sample set of 30 data points, the Gumbel distribution parameters can be estimated using the built in functions in the python scipy library.

`scipy.stats.gumbel_r` calculates the `loc` and `scale` parameters for a given data set where maximum extreme values are required.

`scipy.stats.gumbel_l` calculates the `loc` and `scale` parameters for a given data set where minimum extreme values are required.

These Scipy [2] functions are built into RLC and is selected automatically based on the defined `mode` parameter in each `analysis_criteria` specified.

Fit Tests

Prior to using the calculated Gumbel parameters to determine the probabilistic values, it is important to ensure that the data is a good fit for calculated fit parameters for a Gumbel distribution. Two statistical fit tests are implemented to ensure that the data could have come from the calculated distribution.

Kolomogorov-Smirnov Test

The Kolmogorov–Smirnov (KS) test is a non-parametric method used to determine whether a sample comes from a specific distribution (one-sample test) or whether two samples come from the same distribution (two-sample test). It works by comparing the cumulative distribution functions (CDFs) of the sample(s) and calculating the maximum absolute difference between them, known as the KS statistic. This value reflects how far apart the distributions are, with larger values indicating greater divergence. The KS test is especially useful because it does not assume any particular distribution shape and is sensitive to differences in both location and shape, making it a versatile tool for assessing goodness-of-fit or distributional similarity, particularly for continuous data.

Python's `Scipy` has functions built in to perform the KS test on data. The user can define the KS test significance level (α), however, a typical value of 0.05 is assumed by default.

For each dataset, a KS test is performed versus the calculated Gumbel distribution. If the KS test statistic is greater than the critical value corresponding to the chosen significance level (commonly 0.05), the null hypothesis — that the dataset follows the Gumbel distribution — is rejected. This indicates that the observed data significantly deviates from the expected Gumbel distribution, suggesting a poor fit. Conversely, if the KS statistic is less than or equal to the critical value, the fit is considered statistically acceptable.

If the data is a poor fit the Gumbel distribution should not be used to determine probabilistic values.

Anderson-Darling

Similarly, the Anderson–Darling (AD) test is a statistical test used to evaluate whether a given sample of data follows a specific probability distribution. Like the Kolmogorov–Smirnov test, it compares the empirical cumulative distribution function (CDF) of the sample to the theoretical CDF of the target distribution, but the AD test places more emphasis on the tails of the distribution, making it more sensitive to deviations in those regions.

`Scipy` has functions built in to perform the AD test on data. The user can define the AD test p-value, however, a typical value of 0.05 is assumed by default.

The test produces a statistic that reflects the degree of discrepancy between the sample and the theoretical distribution, and this value is compared against critical values to determine significance. A small p-value (typically less than 0.05) indicates that the sample does not follow the specified distribution, leading to rejection of the null hypothesis. The AD test is especially useful when assessing goodness-of-fit for distributions where tail behavior is important, such as in reliability analysis or extreme value modeling.

If the data is a poor fit the Gumbel distribution should not be used to determine probabilistic values.

Empirical Statistical Evaluation

If either the KS or the AD tests fail, then the Gumbel distribution is not suitable to determine probabilistic values. Therefore, in those cases, the empirical or sample method is used. This approach relies directly on the observed data without assuming any underlying theoretical distribution. Probabilities are estimated by ranking the data and calculating exceedance frequencies based on their position within the sorted dataset. For example, the empirical cumulative distribution function (ECDF) can be used to determine the probability of a value being exceeded, making this method especially useful when the dataset is large and representative, or when distributional assumptions are unreliable.

When using the empirical method, the number of seeds is critical. It is impossible to accurately determine a value with a 1% probability of exceedance if only 10 seeds were run. In general, the required number of seeds can be estimated from the requested probability level.

$$n = 1/(1-p) \tag{8}$$

For a 99% probability level, at least 100 seeds are required.

If the number of seeds run is less than the required number of seeds based on the required probability level **AND** both the KS and AD tests fail, RLC must run more seeds for that case to be able to determine probabilistic values empirically.

If the empirical statistical evaluation is performed, the data goes through a bootstrapping procedure to estimate uncertainty and improve the reliability of probabilistic values. Bootstrapping involves repeatedly resampling the available data with replacement to generate many synthetic datasets, known as bootstrap samples. For each sample, the desired statistic (e.g., exceedance probability) is recalculated, producing a distribution of estimates. This allows for the construction of confidence intervals and quantification of variability, even when the original dataset is limited. Bootstrapping is especially valuable when theoretical assumptions are invalid or when the number of seeds is insufficient for direct inference.

RLC requires at least 1000 bootstrap samples but calculates 10000 by default using the functions built into *Scipy* [2]. This value can be adjusted using the `num_bootstraps` key in the input file.

When all of the bootstrap samples have been created, RLC will determine the probabilistic value with an associated upper or lower confidence interval. By default a 95% confidence interval is used but this value can be modified using the `conf_interval` key in the input file. For maximum cases, it will calculate the upper confidence interval. For minimum cases, it will calculate the lower confidence interval.

Maximum Value Example

A typical distribution of the maximum load from 50 data seeds vs. the best fit Gumbel distribution is shown in Figure 4.

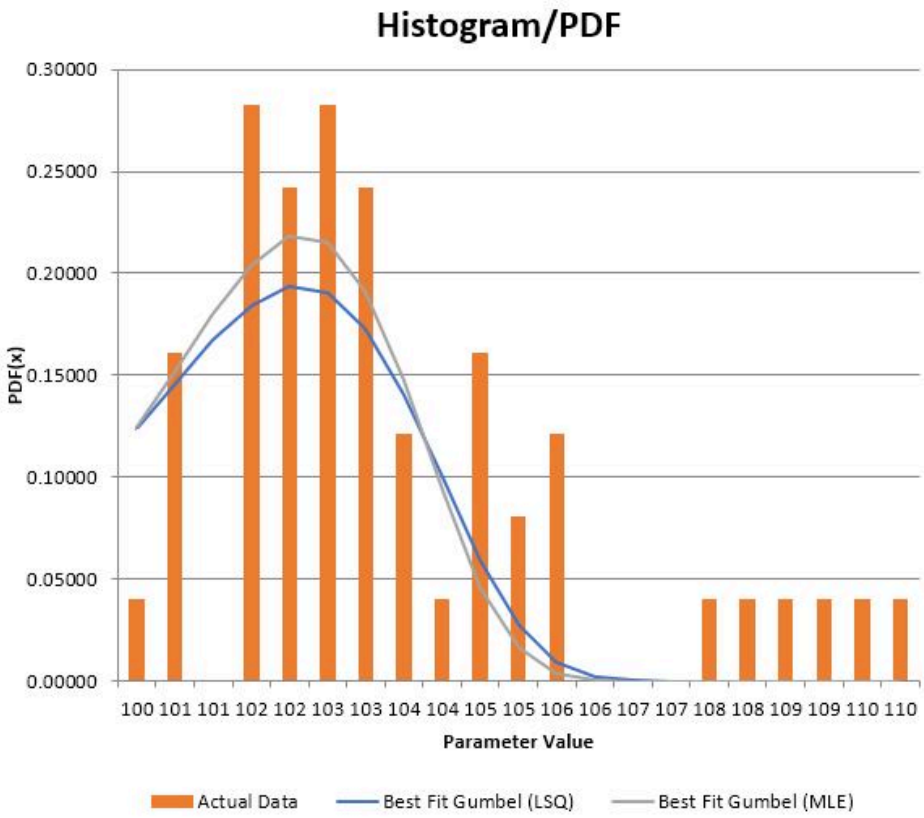


Fig. 4 *Maximum Sample Histogram with Gumbel Curve Fit*

The linear regression of the sample data in Figure 4 is shown in Figure 5.

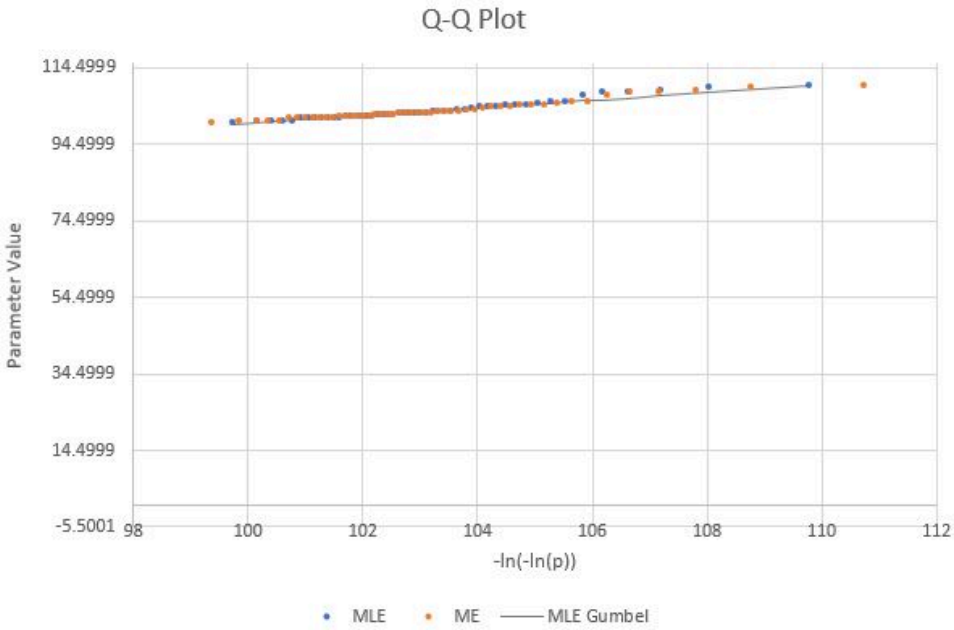


Fig. 5 *Linear Regression of Sample Data* ($\beta = 1.68$; $\mu = 102.25$)

The first step is to perform the KS and the AD test. The KS test provides a measure of how likely the data comes from a population that has a Gumbel distribution. It looks for the largest vertical separation between the empicial CDF and the Gumbel CDF (Figure 6).

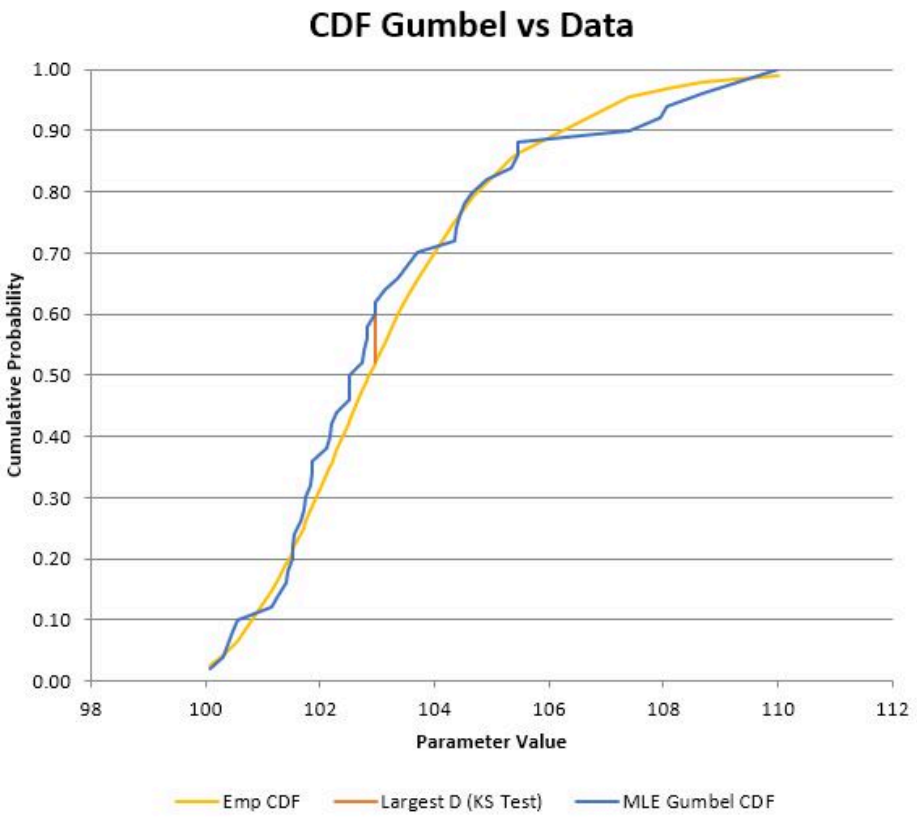


Fig. 6 *Cumulative Distribution Function for Data (Maximum)*

The KS test compares the dataset to a Gumbel distribution and, in this case, returns a statistic of 0.0999. At the 0.05 significance level, the critical value is 0.1923. Since the KS statistic is lower than this threshold, the test passes, meaning there's no significant difference between the data and the Gumbel distribution. The high p-value of 0.700, in this case, further confirms a good fit.

The AD test provides a measure of how likely the data comes from a population that has a Gumbel distribution, but it focuses more on the tails. In this case, the (AD) test results show an adjusted statistic of 0.6012, which is compared against critical values at various significance levels. At the 0.05 level, the critical value is 0.757. Since the adjusted AD statistic is less than this threshold, the test does not reject the null hypothesis, meaning the data is considered a statistically acceptable fit to the Gumbel distribution. The p-value of 0.133 supports this conclusion, indicating no significant deviation between the sample and the theoretical distribution.

Since both the KS and AD test pass, the calculated Gumbel distribution `scale` and `loc` parameters can be used to estimate probabilistic values.

Assuming this data represents the crane load we can use this for checking the crane capacity, using the sample data above, one can determine the load that has a 1/100 probability of occurring (i.e. 99% non-exceedance probability):

$$DHL_{GumbelMax}(0.99) = 102.25 - 1.68 \cdot ln\left(-ln(0.99)\right) \tag{9}$$

$$DHL_{GumbelMax}(0.99) = 110.0kN \tag{10}$$

In other words, in this example, there is a 1% chance that the maximum load during the operation will exceed 110.0 kN.

Minimum Value Example

A typical distribution of the minimum load from 50 data seeds vs. the best fit Gumbel distribution is shown in Figure 7.

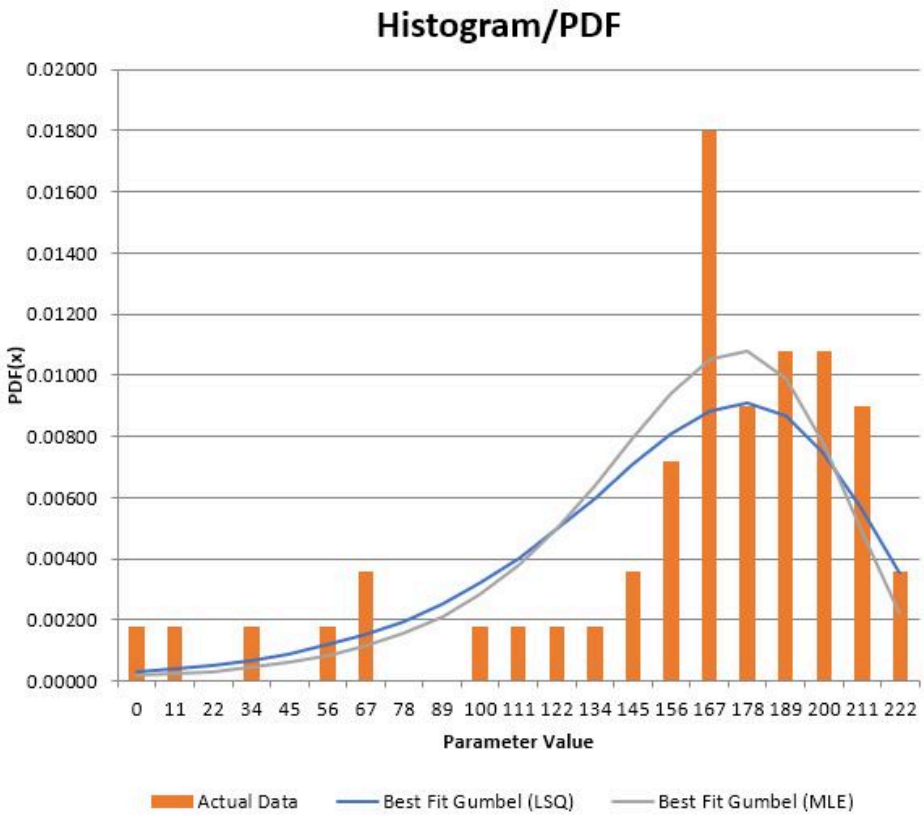


Fig. 7 *Minimum Sample Histogram with Gumbel Curve Fit*

The linear regression of the sample data in Figure 7 is shown in Figure 8.

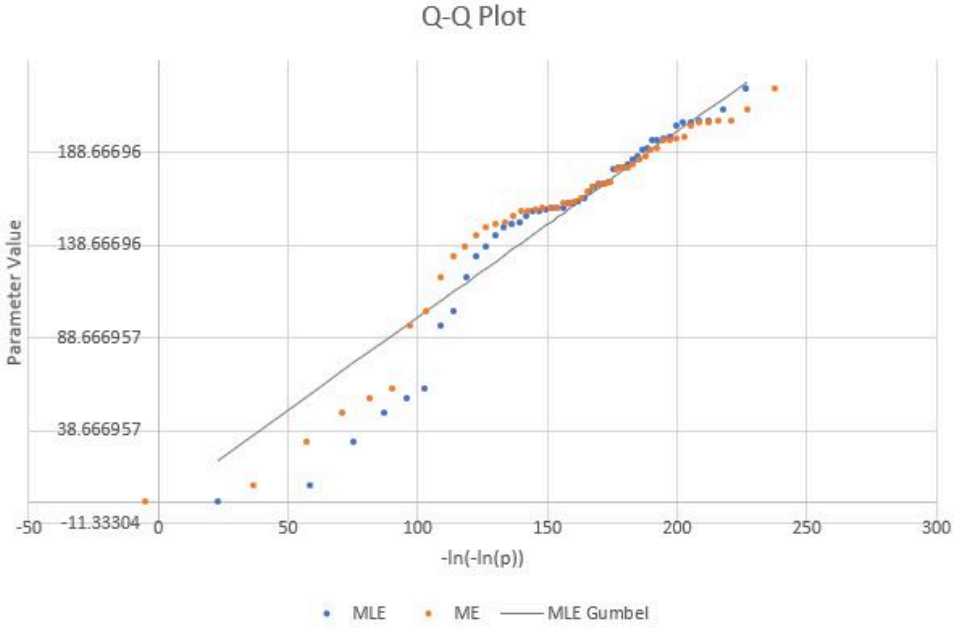


Fig. 8 *Linear Regression of Sample Data* ($\beta = 33.9$; $\mu = 175.7$)

Again, the first step is to perform the KS and the AD test which can be visualized in the CDF plot (Figure 9).

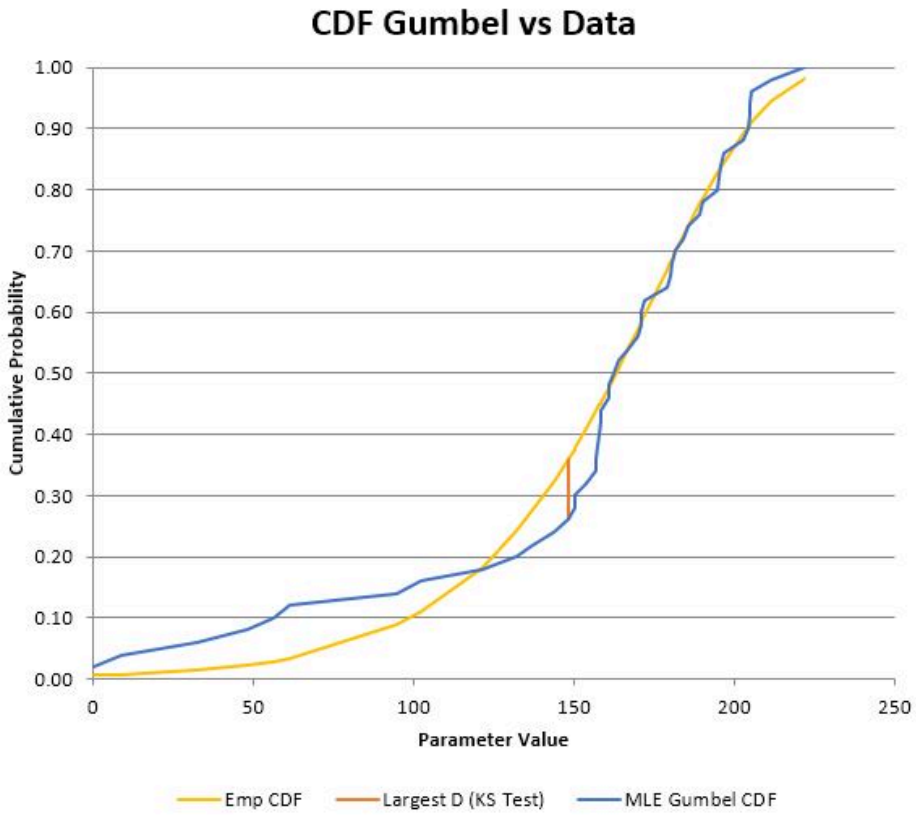


Fig. 9 *Cumulative Distribution Function for Data (Minimum)*

In this case, the KS test returns a statistic of 0.1201. At the 0.05 significance level, the critical value is 0.1923. Since the KS statistic is lower than this threshold, the test passes, meaning there's no significant difference between the data and the Gumbel distribution. The high p-value of 0.467, in this case, further confirms a good fit.

The (AD) test results show an adjusted statistic of 0.9272, which is compared against critical values at various significance levels. At the 0.05 level, the critical value is 0.757. Since the adjusted AD statistic is **greater** than this threshold, the test does reject the null hypothesis, meaning the data is **not** considered a statistically acceptable fit to the Gumbel distribution. The p-value of 0.02 supports this conclusion, indicating a significant deviation between the sample and the theoretical distribution.

Since the AD test fails, the empirical method must be used to determine probabilistic values.

Assuming this data represents the minimum crane load we can use this for checking slack lines in the crane. Using the sample data above, one can determine the load that has a 1/10 probability of occurring (i.e. 90% non- exceedance probability):

For this case, the after bootstrapping the data, the lower 95% confidence level is taken to be 30.3 kN.

In other words, in this example, there is a 10% chance that the minimum load during the operation will fall below 30.3 kN.

Determining the Number of Seeds

Figure 10 shows the effect that the number of seeds analyzed has on the analysis. A number of different structures, headings and zero-up-crossing periods were evaluated. It is clear that with only 10 seeds evaluated a large error will be present in the results, however, as the number of seeds is increased, the resulting error decreases quickly. Based on this, 50 seeds are the recommended as a starting point. If the data does not fit the calculated distribution well and if the required probability levels are larger than 0.98 then additional seeds will be run automatically by RLC.

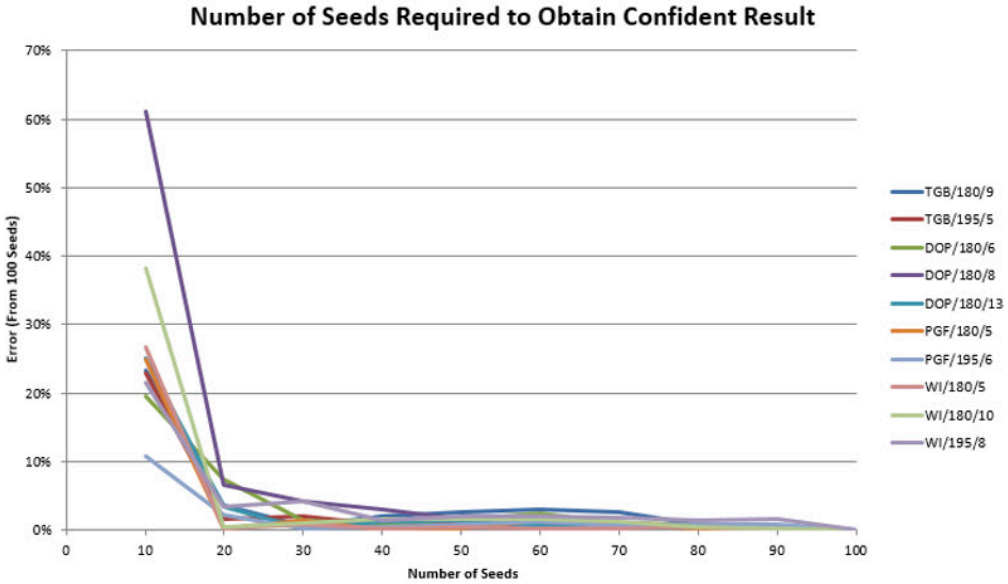


Fig. 10 *Effect of Number of Seeds*

For information, TechnipFMC in Norway also performed testing to determine the effect of the number of seeds analyzed [5].

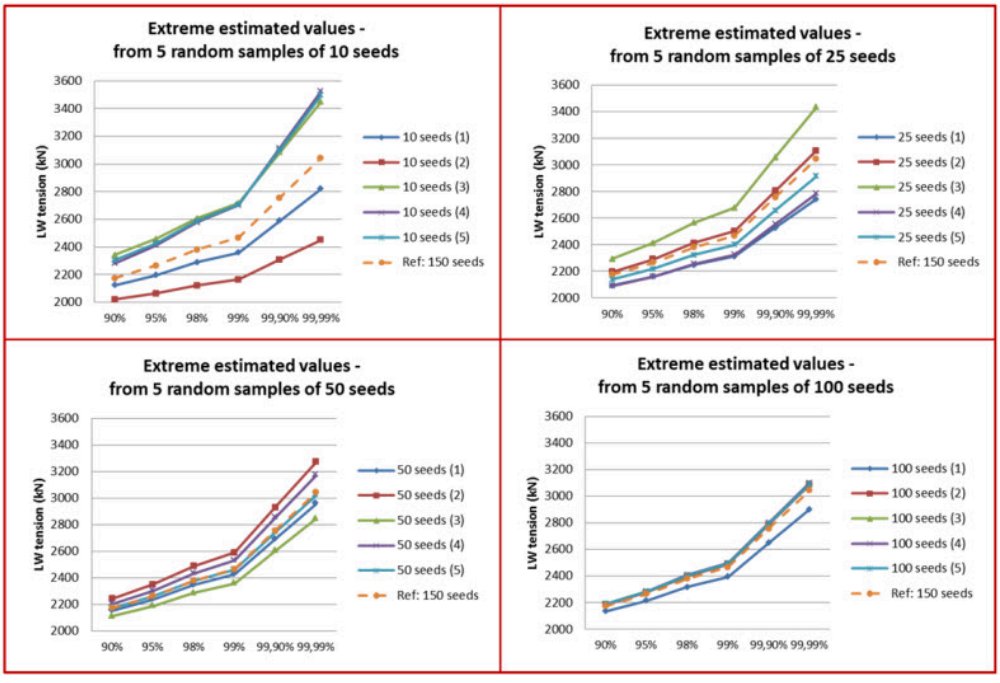


Fig. 11 *Effect of Number of Seeds (Norway)*

Based on this analysis, the following conclusions were made:

- For a 90% probability level of confidence (respectively 99.99%) and using only 10 realizations, the average error is below 1.5% (respectively 3%) whereas the maximum error in the 5 samples is close to 8% (respectively 16%).
- For a 90% probability level of confidence (respectively 99.99%) and using only 25 realizations, the average error is below 1.0% (respectively 1.5%) whereas the maximum error in the 5 samples is close to 6% (respectively 13%).
- For a 90% probability level of confidence (respectively 99.99%) and using only 50 realizations, the average error is below 0.5% (respectively 1.0%) whereas the maximum error in the 5 samples is close to 3% (respectively 8%).
- For a 90% probability level of confidence (respectively 99.99%) and using only 100 realizations, the average error is below 0.2% (respectively 0.5%) whereas the maximum error in the 5 samples is close to 0.5% (respectively 2%).